CPU performance and memory: RAM, ROM and cache

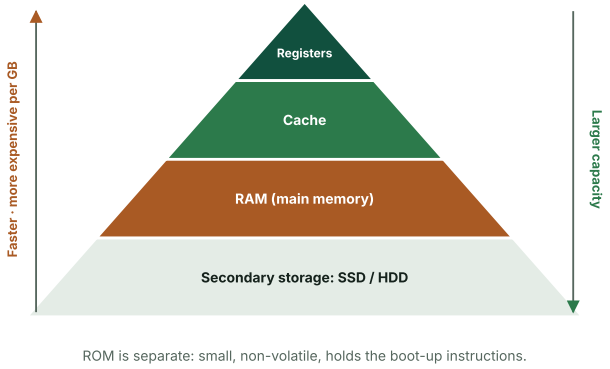
L14 FACT FILE

Specification: students should be able to

- Describe the impact of clock speed, cache size, and number of cores on CPU performance
- Explain the purpose of random access memory (RAM), read only memory (ROM) and cache
- Describe the role of ROM in the boot-up process
- Compare RAM, ROM and cache in terms of volatility, read/write access, size and speed

CCEA fact file: Fact File 11: Computer Hardware

Key words: clock speed, hertz, gigahertz (GHz), core, multi-core, parallel processing, cache, RAM, ROM, volatile, non-volatile, boot program, BIOS, virtual memory, performance



ROM is separate: small, non-volatile, holds the boot-up instructions.

Memory hierarchy pyramid from registers to secondary storage

Three factors that affect CPU performance

The performance of a CPU means how quickly it can carry out instructions. The CCEA specification names three factors. The first is clock speed. The CPU contains a clock that sends out regular pulses, and the clock speed is the number of fetch–execute cycles the CPU can carry out each second. It is measured in hertz, and modern processors are measured in gigahertz: a 3 GHz processor can perform three billion cycles per second. The second factor is the number of cores. A core is a full processing unit: each one has a control unit, an ALU and a set of registers, so a quad-core processor can work on four instructions at the same time. The third factor is cache size. Cache is a small amount of very fast memory close to the CPU that holds frequently used instructions and data. When choosing a computer for a task, such as gaming or editing video, all three should be considered together, along with the amount of RAM.

Why the factors do not simply multiply

A higher clock speed means more instructions are processed each second, but it also produces more heat and uses more battery power. Also, if the other parts of the computer cannot supply data quickly enough, the CPU ends up waiting, so doubling the clock speed does not double the overall speed. In the same way, more cores only help if the software has been written to split its work into parts that can run in parallel. A simple program that does one thing at a time may use only one core, and the others sit idle. More cores are very useful when several programs are open, or for jobs that split up easily, such as rendering video. A larger cache means the CPU finds what it needs in the cache more often. Each time it does, it avoids a slower trip to RAM, so processing is faster. However, cache is expensive to make, which is why it is kept small.

RAM, ROM and cache

Internal memory is memory the CPU can access directly. Random access memory (RAM) is the main memory. Its purpose is to hold the operating system, programs and data that are currently in use, such as a web browser and the page you are reading. RAM can be read from and written to, and its contents change all the time. It is volatile, meaning everything in it is lost when the power is switched off, which is why files must be saved to storage. Read only memory (ROM) can be read but not normally changed. It is non-volatile, so it keeps its contents without power. Its purpose is to store the boot program, the instructions the computer needs as soon as it is switched on. Cache is the third type. Like RAM it can be read from and written to and it is volatile, but it is much smaller and much faster, and it sits inside or right beside the CPU.

Booting up and running out of RAM

When you press the power button, RAM is empty, so the computer cannot start from it. Instead, the CPU runs the start-up instructions stored in ROM, often called the BIOS. These instructions check that key hardware is working and then load the operating system from the hard disc or SSD into RAM. Once the operating system is in RAM, it takes over. While the computer is running, RAM can fill up if many programs are open. The operating system then uses part of the storage drive as extra memory, called virtual memory, moving data that is not being used right now out of RAM. Because storage drives are much slower than RAM, the computer slows down as data is swapped back and forth. This is why adding more RAM can make a computer faster: it reduces the need for virtual memory. The textbook says larger RAM makes the processor faster; more precisely, it stops the processor having to wait for slow storage.

Key terms from the mark schemes

Clock speed
The number of fetch–execute cycles a CPU can carry out per second, measured in hertz (e.g. 3 GHz = 3 billion cycles per second).
`Credited: Processing cycles per second`

Cache
Small, very fast memory on or near the CPU that holds frequently used instructions and data, so they need not be fetched from slower RAM.
`Credited: Cache memory`

ROM
Read only memory: non-volatile memory whose contents cannot normally be changed, used to store the boot-up instructions (BIOS).

Non-volatile
Describes memory or storage that keeps its contents without power, such as ROM and SSDs.

Virtual memory
Part of a storage drive used as extra memory when RAM is full; it is much slower than RAM.

Core
An independent processing unit inside a CPU; each core has a control unit, an ALU and registers, so it can process instructions at the same time as other cores.
`Credited: Processor core`

RAM
Random access memory: volatile read/write main memory that holds the operating system, programs and data currently in use.
`Credited: Main memory`

Volatile
Describes memory that loses its contents when the power is switched off, such as RAM and cache.

Boot program
Instructions stored in ROM that run when the computer is switched on, checking hardware and loading the operating system into RAM.
`Credited: BIOS; bootstrap loader`

Multi-core processor
A CPU that contains two or more cores, so it can process several instructions in parallel.
`Credited: Dual-core / quad-core`

Exam focus

Performance and memory questions appear often. Clock speed and its units came up in 2021 Q5 and 2022 Q12(c). RAM and ROM have been tested in 2019 Q5(a) and 2024 Q6(e), and ROM's role in booting appeared in 2025 Q4. Cache came up in 2024 Q5(c) and as a matching term in 2025 Q13. For 'describe the impact' questions, give the cause and the effect: 'a higher clock speed means more fetch–execute cycles per second, so more instructions are processed per second'. For cache, the marks usually need two ideas: it is faster to access than RAM, and it holds frequently used data or instructions. For RAM versus ROM, give a matched pair of differences, such as 'RAM is volatile but ROM is non-volatile', rather than describing only one memory. Common mistakes are saying clock speed is measured in bytes, saying ROM holds the user's files, and saying that cache is part of the hard disc.

Unit 1 2025 Q13	[10]	Term matching: hardware, memory, networking, malware
Unit 1 2025 Q4	[7]	Boot process, OS roles, storage conversion
Unit 1 2024 Q6	[13]	Network statements, bus disadvantages, LAN, encryption, HTTPS, RAM vs ROM, malware routes
Unit 1 2024 Q5	[10]	Spreadsheet functions, CPU purpose and components, immediate-access store
Unit 1 2022 Q12	[7]	WAN, encryption, firewall, processor performance

? Check yourself

1. State the unit used to measure clock speed. [1]
2. Describe the impact of increasing the clock speed on the performance of a CPU. [2]
3. A computer in the school library has a 3.2 GHz processor. State how many cycles per second this is, written in full. [1]

Answers: 1. Hertz / Hz / gigahertz (GHz) (1). 2. Higher clock speed means more fetch–execute cycles per second (1) / so more instructions are processed per second and performance is faster (1). 3. 3,200,000,000 (three thousand two hundred million / 3.2 billion) cycles per second (1).